Where Should Layers Count? Residual Transport Weighting for Neural Spectral Capacity

CS5491 Artificial Intelligence — Project Report

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1 Problem and motivation

Neural Spectral Capacity (NSC) [1] scores an architecture from its specification alone, $S_0 = \sum_l [3H_l \psi_{\text{MP}}(d, d/H_l) + \psi_{\text{MP}}(d, d) + 2\psi_{\text{MP}}(d, f_l)]$, where $\psi_{\text{MP}}(m, n, \sigma)$ is the Marchenko–Pastur estimate of $\ln\det(I + W^\top W)$ for $\sigma = 0.02$. Within a matrix this has a channel interpretation, but the equal sum across layers is a modelling choice: reordering layers cannot change S_0, and the starter's l/L weighting breaks such ties without a mechanism. We ask whether *residual structure determines how much each layer should count, and whether the answer depends on where LayerNorm (LN) is placed*. All 200 archived LiteTransformerSearch [2] models are Post-LN (`pre_lnorm=false`), so the placement decides what theory predicts for this panel. We contribute a specification-only derivation of per-sublayer transport weights for three placements, building on residual Jacobian products [3] and DeepNet's Post-LN analysis [4]; its Monte Carlo (MC) validation; and a frozen weighted score evaluated against its ablation and baselines. NSC and the evaluator are reproduced from the course starter.

2 Method

Reference model. Sublayers $s = 1, \ldots, 2L$ alternate attention and FFN. Post-LN updates $x \leftarrow \text{LN}(x + F_s(x))$, Pre-LN $x \leftarrow x + F_s(\text{LN}(x))$ with a final LN, and "none" $x \leftarrow x + F_s(x)$. Weights are i.i.d. $\mathcal{N}(0, \sigma^2)$ with NSC's $\sigma = 0.02$; biases are zero, LN gains one, no dropout. At this scale attention logits are near zero, so attention is uniform causal averaging over $K_i = \text{mem} + i$ keys followed by $W_V W_O$; memory enters the forward pass but is detached in the backward pass, as in Transformer-XL [9]. FFN is $W_2\,\text{ReLU}(W_1 x)$. Input tokens are i.i.d.; $n = \text{mem} = 192$ as in the archive.

Score and ablation. With u_s the branch output of sublayer s, y the network output and δ an i.i.d. unit perturbation, the transport factor and the score are

$$T_s = \tfrac{1}{nd}\,\mathbb{E}\big\|(\partial y/\partial u_s)\,\delta\big\|^2, \qquad S = \textstyle\sum_s T_s \Psi_s,$$

where Ψ_s splits NSC by sublayer: $3H\psi_{\text{MP}}(d, d/H) + \psi_{\text{MP}}(d, d)$ for attention and $2\psi_{\text{MP}}(d, f)$ for FFN. The ablation $T_s \equiv 1$ is exactly S_0. As $\sigma \to 0$, $T_s \to 1 - 2/d$ (with LN) or 1 (none), so S becomes a constant multiple of NSC.

Mean-field recursion. We track per-token energy E and mean pairwise token correlation ρ. With branch gains $a = d^2\sigma^4$ (attention) and $b = fd\sigma^4/2$ (FFN), a unit-energy input produces output energy m and pairwise product p: $m = a[c_1 + \rho(1 - c_1)]$, $p = a[c_2 + \rho(1 - c_2)]$ for attention and $m = b$, $p = 2bk_1(\rho)$ for FFN, where c_1, c_2 are causal averages of $1/K$ and k_0, k_1 are ReLU arc-cosine kernels [8]. In Post-LN, $\rho \leftarrow (\rho + p)/(1 + m)$ and LN resets $E = 1$. Backward, a perturbation from s starts with energy $\nu = 1/(1 + m_s)$ and pairwise product $q = 0$. Each later attention applies $\nu \mathrel{+}= a(A_1\nu + A_2 q)$, $q \mathrel{+}= a(B_1\nu + B_2 q)$, whose coefficients count only current, non-memory tokens; each FFN applies $\nu \mathrel{*}= 1 + b$, $q \mathrel{*}= 1 + 2bk_0(\rho_t)$; each LN divides both by $1 + m_t$. Finally $T_s = (1 - 2/d)\,\nu$. Pre-LN and none use the same quantities without the per-step division.

Why Post-LN nearly cancels. Each later sublayer multiplies the transported energy by $(1 + g_t)/(1 + m_t)$: its Jacobian gain over the forward gain that its LN divides out. For FFN, $g_t = m_t$ exactly. For attention, $g_t < m_t$, because correlated tokens (the onset of rank collapse [6,7]) add

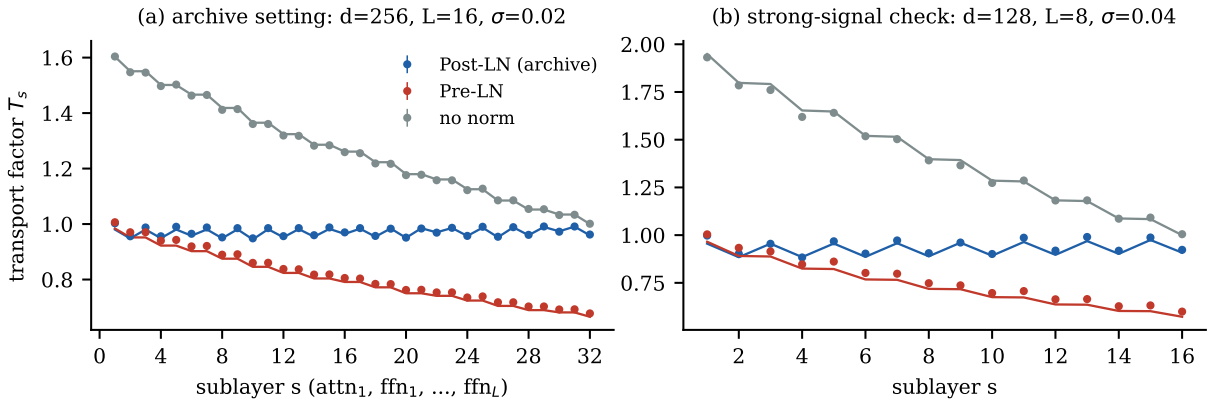


Figure 1: Predicted T_s (lines) and MC measurements (dots). (a) Archive setting, one $d = 256$, $L = 16$ development model, 10 seeds, ± 2 s.e. (b) Strong-signal check outside the archive ($\sigma = 0.04$).

Table 1: MC validation at $n = \text{mem} = 192$: 28 development models, 532 sublayers per placement.

Placement	RMS error (95% CI)	excl. first	MC s.e.	profile corr.	max/min
Post-LN (archive)	0.81% (0.64–0.96)	0.45%	0.33%	0.92	1.06
Pre-LN	2.80% (2.16–3.40)	2.79%	0.41%	0.999	1.15
none	0.24% (0.21–0.28)	0.24%	0.20%	0.999	1.15

common-mode energy forward that a perturbation does not see, and memory contributes only forward. Hence $T_s \approx (1 - 2/d)/(1 + m_s)$ times a product slightly below one. The weight depends mainly on the sublayer's own width, and earlier sublayers are transported slightly less, in line with larger gradients near the output in Post-LN [4,5]. Without the division (Pre-LN, none), $\prod_{t>s}(1+g_t/E_t)$ grows towards the input and early layers dominate, which is the opposite of l/L.

3 Validation of the derivation (no labels used)

We built random residual networks with full softmax attention, memory from two warm-up segments and finite-difference Jacobian probes, for 28 development specifications (depth quantiles within each width) with 10 seeds and 2 probes (Fig. 1a, Table 1; medians for s.e., profile correlation and max/min). In Post-LN the error is close to MC noise once the first sublayer is excluded. That sublayer is biased by about $3/d$ because the reference feeds unnormalized $\mathcal{N}(0, 1)$ input to the first LN; in Pre-LN this raw input persists in the stream, so the final LN carries the same offset to every sublayer, changing scale but not shape. An earlier i.i.d.-token version had 3.3% RMS error (10% maximum), which motivated the correlation term. In a strong-signal regime ($\sigma = 0.03$–0.05, Fig. 1b) errors stay at 0.7–4%, while measured weight spreads reach $3.2\times$ without normalization but only 1.1–$1.25\times$ in Post-LN.

For the starter's reorder pair ($d = 128$, FFN widths $[256, 512]$ vs. $[512, 256]$, tied at 121.820 by NSC), the Post-LN score gives 119.6127 vs. 119.6123, barely breaking the tie. Pre-LN (119.43 vs. 119.36) and none (122.28 vs. 122.21) favour the wider early layer, whereas l/L (96.96 vs. 85.77) favours the opposite. On the development panel, Post-LN weights span 0.84–1.00, with a median within-model max/min of 1.025; Pre-LN weights span 0.49–1.00.

4 Ranking and selection

Protocol. All choices used the 100 development architectures (validation PPL). We then froze the method (git tag `v2-frozen`) and evaluated the 100 final architectures (test PPL) once. Be-

Table 2: Rank correlation with negative PPL; ctl = size-controlled pairs; flips = final pairs ordered differently from NSC (all / controlled). Only depth and FLOPs have score ties.

	Development			Final (held out)			
Score	ρ	τ_b	ctl τ_b	ρ	τ_b	ctl τ_b	flips
Ours (Post-LN)	0.970	0.861	0.622	0.967	0.844	0.330	5 / 0
Ablation = NSC	0.970	0.861	0.622	0.966	0.843	0.330	–
l/L example	0.970	0.865	0.622	0.969	0.853	0.319	97 / 11
Pre-LN weights	0.971	0.865	0.644	0.967	0.845	0.330	9 / 2
Projection params	0.970	0.861	0.600	0.964	0.841	0.308	23 / 4
Reference FLOPs	0.968	0.859	0.622	0.965	0.842	0.330	22 / 2
Total params	0.728	0.576	0.133	0.742	0.590	0.276	1134 / 57
Depth	0.479	0.360	0.652	0.485	0.362	0.379	1238 / 6

Table 3: Final-split selection: mean regret in PPL (best selected minus best feasible) over 200 tie-break or random trials. Pool sizes 25/57/79/96.

Cap / k	10M/1	10M/3	20M/1	20M/3	30M/1	30M/3	50M/1	50M/3
Ours, NSC, l/L, FLOPs	0	0	1.14	0.85	1.62	0.77	1.37	1.33
Projection params	0	0	0.85	0.85	1.62	0.77	1.37	1.33
Total params	3.72	3.72	8.44	1.49	0	0	1.13	1.13
Depth	14.42	8.32	15.44	4.26	3.45	0.00	3.46	1.60
Random	12.18	3.97	15.36	6.29	15.53	5.93	15.36	5.85

sides the required baselines, we report mismatched Pre-LN weights, depth and projection parameters $\sum_l (4d^2 + 2df_l)$. Size-controlled pairs share d and have a total-parameter ratio ≥ 0.90 (180 development, 185 final; they overlap and are not independent).

Results. The weighted score and its ablation are indistinguishable (Tables 2, 3): final ρ is 0.9667 versus 0.9659, and selection is identical in all eight cap/k cells on both splits. Across both splits only 7 pairs change order relative to NSC (5 become correct, 2 wrong). All are near-ties within 1.2%, and 6 compare different widths, so they reflect the width-dependent scale $(1-2/d)/(1+m_s)$ rather than layer position. Pre-LN weights also give the same final selection, so neither weighting helps. On development, NSC's only non-zero 10M regret (1.94 at $k = 1$) came from one top-1 swap and did not recur on the final split.

5 Discussion and limitations

What the score learns beyond size. Very little on this panel. NSC is almost rank-identical to projection parameters (development Spearman 0.9997). Its advantage over total parameters comes from excluding embeddings: within $d = 64$, ρ is 0.98 versus 0.09 on development and 0.92 versus 0.18 on final. Post-LN transport is nearly constant within a model, so it cannot add much either. The derived weights would matter where normalization does not divide out residual growth, i.e. Pre-LN models, which this panel lacks; testing that needs new trained outcomes. The result is a supported negative one: under Post-LN and small initialization, equal summation is the first-order aggregation implied by residual structure.

Instability of size control. For every score, controlled τ_b falls from about 0.62 on development to about 0.33 on final. This is a property of the panel's near-size pairs, not of any method.

Scope. T_s describes signal propagation at initialization, not learning: a branch with zero output can still learn, which we verified on a small example. The model omits relative position atten-

tion, adaptive embeddings, dropout and softmax non-uniformity, and assumes i.i.d. inputs. The archive has one outcome per architecture, so reported variability covers only tie-breaking and random selection, not training seeds.

Reproducibility and LLM use. The repository README lists the environment and exact commands; the frozen method is tag `v2-frozen`. Scoring 100 architectures takes 0.12 s cold and 5 ms warm (NSC: 0.12 s and 0.4 ms) on an Intel Core Ultra 9 290HX; the MC validation takes 27 min on 22 processes. Claude (Anthropic) and ChatGPT/Codex (OpenAI) assisted with literature reading, code and drafting. All numbers come from the repository's scripts, and the team is responsible for the content.

References

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